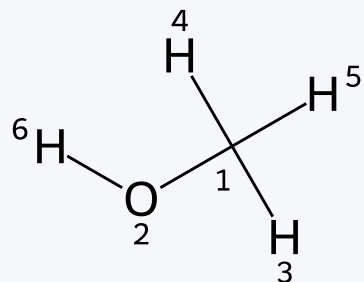


a One endpoint pair, two GRAFT bond-event hypotheses

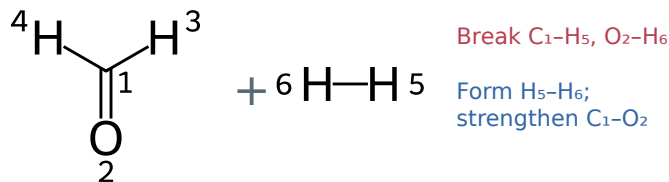
Methanol → formaldehyde + H₂ | All H atoms explicit; numbers track reactant identities



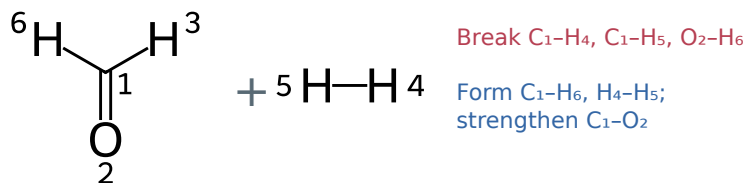
Same starting atoms



A · 4 changes

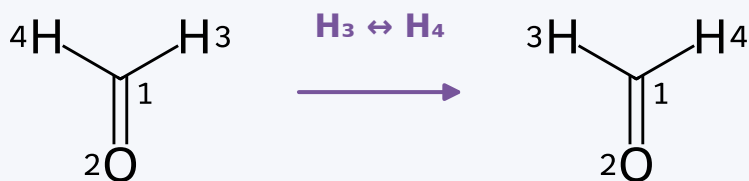


B · 6 changes



b A representative still has allowed shuffles

Class A: exchange the two retained methyl H atoms



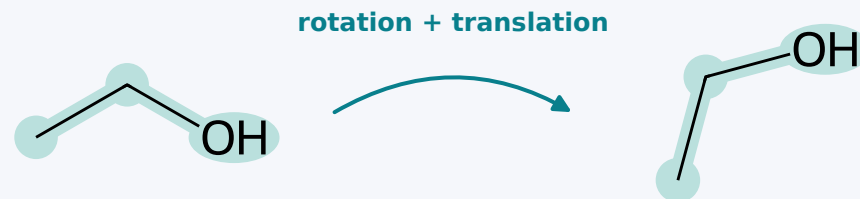
H₅-H₆ is unchanged

Same-class query: allowed

A → B: forbidden in A; allowed across saved classes

c Fragments can guide geometric preparation

Illustrative retained fragment: move C-C-O together



Keep internal bonds; refine reactive contacts

Then test a transition-state guess

Proposed use · no barrier or pathway validation here